The Collapse of Phase Space: Narrative as an Operational Thermodynamic Quantity in Inference Pruning

Constraint Density, Realization Entropy, and Structural Friction in Autoregressive Language Models

Abstract

Contemporary optimization of systems built on Large Language Models has largely concentrated on two layers: reducing the number of input tokens and imposing syntactic constraints on outputs through schemas, grammars, or structured formats. These techniques are important, but they do not exhaust the problem. A request may be syntactically well specified while remaining semantically underdetermined, forcing the model to independently resolve the intended objective, scope, level of detail, discursive register, relevant exceptions, and terminal form of the response.

This paper develops the hypothesis of **Semantic Thermodynamics**, according to which a coherent narrative instruction acts as a boundary condition on the output distribution of a probabilistic model. Narrative, in this framework, does not refer to literary ornamentation, but to the coordinated specification of operational identity, objective, scope, exclusions, output form, and failure state. Such structure reduces the effective volume of output trajectories compatible with a request and may convert a modest increase in input cost into a substantially larger reduction in generation cost.

A reanalysis of Experiment Zero, comprising 100 calls organized into 50 control-treatment pairs, found a 44.44% increase in input tokens but a 79.29% reduction in completion tokens, a 46.99% reduction in total tokens, and a 60.73% reduction in mean end-to-end latency. The 50 control calls produced 50 distinct textual realizations, whereas all 50 optimized calls converged to a single realization. Experiment One, consisting of 120 calls distributed across four levels of narrative density, revealed a U-shaped cost curve: minimum latency and token expenditure occurred not under the most elaborate formulation, but under a light-gravity condition. Extreme overload increased mean latency to 6.775 seconds and mean completion-token consumption to 752.07 tokens.

These results support an operational formulation of the **Law of Entropic Proportionality**: for a fixed model, serving regime, and quality threshold, there exists an optimal density of semantic constraint, and that density depends on the ambiguity and complexity of the underlying task. Coherent constraints below that point compress the effective output space; redundant, conflicting, or disproportionate constraints above it generate **Structural Friction**. Narrative therefore emerges as a semantic control layer complementary to, rather than competitive with, syntactic decoding constraints.

Keywords: large language models; prompt engineering; semantic entropy; inference latency; reasoning tokens; narrative constraints; autoregressive inference; AI infrastructure optimization.

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1. Introduction: The Hidden Cost of Underdetermination

A production system can guarantee that an answer is valid JSON without guaranteeing that the model knows **which information deserves to occupy that JSON**. It can require a property named `penalty` without fully resolving whether the value should be nominal or inflation-adjusted, whether the currency should be preserved, whether the condition triggering the penalty should also be included, or whether missing information should result in `null`, an estimate, or a refusal.

A syntactic constraint determines the geometry of the container. It does not fully determine the semantic selection of its contents.

Constrained decoding and structured-output methods can formally restrict the set of admissible tokens, guaranteeing conformity with grammars or schemas. Prior research has shown that such techniques can improve syntactic correctness and, in some settings, semantic performance as well. Other results, however, indicate that excessively rigid structures may push models toward lower-confidence sequences and degrade performance in certain generative tasks. Structure therefore has both a benefit and a cost; its usefulness depends on the relationship between the imposed constraint and the distribution the model has learned to generate.

The problem investigated here precedes syntactic validation. It concerns the amount of indeterminacy that remains when a model receives a task.

In the absence of clear boundaries, even a seemingly simple instruction contains multiple unstated questions:

Should the model merely answer, or explain?

Should it include caveats?

Which professional or conceptual perspective should govern interpretation?

Which portions of the input are relevant?

Which plausible alternatives should be discarded?

What should happen if the necessary information is absent?

Every decision left unspecified remains a degree of freedom in generation.

The foundational formulation of Semantic Thermodynamics organizes these decisions into six components: Persona (\Psi), Objective (\vec{\Omega}), Scope Boundary (\oint), Negative

Constraints ($\bar X$), Output Matrix (M), and Ground State (S_0). The original hypothesis proposed that their coherent combination could reduce narrative entropy and the inferential work associated with a request.

The central hypothesis of the present paper is more precise:

> **A structured operational narrative modifies the conditional output distribution of a language model. When that structure removes relevant degrees of freedom without introducing conflicts or auxiliary tasks, the work saved during generation can exceed the additional work introduced during prompt processing.**

This proposition does not require the assumption that a model understands a narrative in the same way a human does. It requires only the established fact that context modifies the probabilities assigned to subsequent tokens.

An autoregressive model produces a sequence ($Y=(y_1,\ldots,y_T)$) according to a conditional distribution of the form

$$\prod_{t=1}^{T} p_\theta(y_t\mid y_{<t},X,P),$$
]

where (X) represents the information being processed and (P) represents the instruction.

Changing (P) changes the internal states generated by the Transformer and, consequently, the probability distribution computed at each decoding step.

Narrative therefore does not need to make individual words “process faster.” Its potential effect is to make entire classes of continuation less probable before they materialize either as visible output or as non-visible generated computation.

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2. What Narrative Means in This Framework

In ordinary language, narrative usually refers to story, plot, or dramatic construction. Within Semantic Thermodynamics, the term has a broader and more technical meaning.

A narrative is the structure that jointly answers six fundamental questions:

Who is operating?

Toward which terminal state?

Within which boundaries?

Which trajectories are forbidden?

What form constitutes a valid conclusion?

In which state should the system settle when no valid solution exists?

This architecture converts an open-ended request into a topology of action.

2.1 Persona (\Psi): Policy Selection

The Persona does not literally deactivate parameters or isolate a specialized physical subnetwork. Instead, it acts as a policy-selection mechanism: it conditions the model toward patterns of interpretation and response associated during training with a particular functional role.

“Forensic financial auditor” and “general conversational assistant” induce different priors regarding relevance, ambiguity tolerance, evidentiary standards, and output style.

Instruction-tuned models are explicitly trained to condition behavior on descriptions of user intention and task context. Persona is therefore best understood not as theatrical decoration, but as an initial semantic constraint on behavioral policy.

2.2 Teleological Vector (\vec{\Omega}): Terminal-State Definition

The Teleological Vector defines the desired endpoint.

An instruction such as:

> “Analyze the contract.”

admits a very large number of valid outcomes.

By contrast:

> “Extract exclusively the amount of the early-termination penalty.”

contracts the space of acceptable outcomes toward a single functional coordinate.

2.3 Scope Boundary (\oint): Evidentiary Topology

The Scope Boundary defines which evidence may participate in the decision.

It distinguishes information that is semantically available from information that is operationally authorized.

A fact can exist in the context without belonging to the task.

This distinction becomes increasingly important in long-context systems, where the availability of information does not imply its relevance.

2.4 Negative Constraints (\bar X): Destructive Pruning

Negative constraints explicitly eliminate classes of continuation that would otherwise remain plausible under the model's default behavior.

“Do not explain.”

“Do not evaluate jurisdiction.”

“Ignore greetings.”

These instructions do not erase already processed input tokens. Instead, they reduce the legitimacy of certain semantic directions as destinations of generation.

2.5 Output Matrix (M): Realization Crystallization

The Output Matrix defines the terminal form of the response.

Its effect is not merely aesthetic.

Requiring a single word, a categorical label, or an object with predetermined fields reduces **realization entropy**: the number of alternative surface forms through which the same semantic content could be expressed.

2.6 Ground State (S_0): Safe Terminal Absorption

The Ground State specifies a valid low-energy outcome for cases in which the requested information cannot be recovered.

Without an explicit fallback, a model may interpret the production of *some* answer as more important than fidelity to the source.

Outputs such as

`NOT_FOUND`

or

`null`

create an absorbing terminal state when the evidence required by the objective does not exist.

These six components are not equally necessary for every request.

That observation is the starting point of Experiment One.

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3. Semantic Phase Space: From Metaphor to Operational Definition

The term *phase space* should not be interpreted as claiming that a language model explicitly constructs and traverses a complete tree of all possible future texts.

During inference, a Transformer computes representations and a distribution over the next token. It does not explicitly simulate every possible complete continuation before selecting one.

The relevant space is therefore the **effective set of trajectories to which the model assigns non-negligible probability**.

We may represent this set as

$$
\left\{
Y:
p_\theta(Y\mid X,P)>\varepsilon
\right\},
$$

for some operational threshold (ε).

The effective volume of this space can be approximated as

$$
\exp
\left[
H_\theta(Y\mid X,P)
\right],
$$

where (H_θ) denotes the entropy of the output distribution.

The more probability mass is concentrated into a small number of trajectories, the smaller this effective volume becomes.

Yet output entropy contains distinct phenomena.

Let (A) represent the semantic answer class—for example, *the safe door is blue*—and let (Y) denote the concrete sequence of text used to communicate that answer.

When (A) can be determined from (Y), output uncertainty can be decomposed as

$$
H(A\mid X,P)
+
H(Y\mid A,X,P).
$$

The first term is **answer entropy**: uncertainty regarding what semantic answer will be produced.

The second is **realization entropy**: uncertainty regarding how that same answer will be expressed.

This distinction is fundamental.

Fifty different responses may communicate exactly the same proposition. In such a case, lexical diversity is high while semantic answer entropy is low.

Research on semantic entropy in LLMs has demonstrated precisely why outputs must be grouped by meaning before textual diversity can be interpreted as substantive uncertainty.

Semantic Thermodynamics may therefore produce at least three distinct effects:

1. reduction in uncertainty over **which material answer** should be produced;
2. reduction in the number of **discursive realizations** through which that answer is expressed;
3. reduction in the amount of **visible and non-visible generation** required before termination.

The experiments presented here strongly affect the second and third components. Evidence concerning the first is more limited because the underlying material answers were already highly constrained.

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4. From Entropy to Infrastructure Cost

The latency of a request to a remotely served language model is not equivalent to pure GPU execution time.

It may include, among other components:

- network transmission;
- queueing;
- prompt processing;
- model generation;
- response serialization and return.

The experimental scripts measured the entire interval between request dispatch and receipt of the final API response. The observed quantity must therefore be described as **end-to-end latency**, not direct hardware execution time.

Autoregressive systems also exhibit an important asymmetry between prompt processing and token generation.

During the *prefill* phase, input tokens can be processed with substantial parallelism.

During *decode*, output tokens are generated sequentially.

For this reason, the number of generated tokens frequently exerts a stronger influence on total request latency than a modest increase in prompt length.

A useful operational cost function for a prompt (P) can therefore be written as

c_p T_p(P)
+
c_c,\mathbb{E}[T_c\mid P]
+
c_\ell,\mathbb{E}[L\mid P]
+
c_e,\mathbb{E}[\mathcal{E}_{\text{task}}\mid P],
]

where:

- (T_p) is the number of input tokens;
- (T_c) is the reported number of completion tokens;
- (L) is end-to-end latency;
- (\mathcal{E}_{\text{task}}) is material or structural error;
- (c_p,c_c,c_\ell,c_e) are economic or operational weighting coefficients.

This formulation makes one point explicit:

> **The shortest prompt is not necessarily the cheapest prompt.**

An increase in input length is economically favorable whenever it produces a larger and more valuable contraction in generation, latency, or error.

An additional interpretive issue concerns `completion_tokens`.

In reasoning-capable models, this quantity may include internally generated tokens that are not exposed in the visible answer, in addition to formatting or delimiter tokens not apparent in the final content.

The visible response may therefore represent only a fraction of the generated work recorded by API telemetry.

This pattern is visible in the experiments.

A final answer containing only the word *Blue* consumed an average of 63.32 completion tokens.

A JSON response containing only a small number of visible characters consumed an average of 40.83 completion tokens.

Without a decomposition of `completion_tokens_details`, the excess cannot be attributed exclusively to hidden reasoning. It can, however, be stated that visible text accounts for only part of the generation measured by the API.

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5. Experiment Zero: Semantic Contraction and Output Crystallization

5.1 Experimental Design

Experiment Zero compared two prompts intended to solve the same logical problem.

The control prompt presented three doors and requested an analysis.

The optimized prompt added an operational identity, a high-stakes narrative frame, more direct rules, an explicit fallback, and a one-word output matrix.

The experiment executed 50 iterations, with one control and one optimized call in every iteration, resulting in 100 total requests.

The two conditions were launched asynchronously, while the order of task creation alternated between iterations to reduce systematic server-delay bias.

For every response, the script recorded:

- prompt tokens;
- completion tokens;
- end-to-end latency;
- final answer.

5.2 Results

Condition	Calls	Prompt Tokens	Completion Tokens	Total Tokens	Mean Latency	Unique Textual Outputs
Control	50	108.00	305.70	413.70	3,489.38 ms	50
Optimized	50	156.00	63.32	219.32	1,370.39 ms	1
Change	—	**+44.44%**	**−79.29%**	**−46.99%**	**−60.73%**	**−98.00%**

In absolute terms, the optimized formulation added 48 prompt tokens while saving an average of 242.38 completion tokens.

This produces an exploratory **semantic leverage ratio**

$$-\frac{\Delta T_c}{\Delta T_p}$$

5.05,
]

or approximately **5.05 completion tokens avoided for every additional input token**.

The net saving was 194.38 tokens per request.

Paired resampling of the 50 experimental blocks produced approximate 95% intervals of −81.31% to −76.99% for the change in completion tokens and −63.64% to −57.63% for latency.

The observed contraction was therefore not driven by a small number of extreme measurements.

5.3 What Actually Collapsed?

All 100 responses identified the blue door.

The experiment therefore did not transform incorrect answers into correct ones.

Within the observed sample, semantic answer entropy (A) was approximately zero under both conditions.

The difference appeared in realization.

All 50 control responses were textually distinct. Many introduced caveats regarding color models, explained why green could be considered primary under RGB conventions, or justified the interpretive assumptions used to resolve the problem.

The empirical entropy of exact output strings was

\log_2 50
\approx
5.64\ \text{bits}.
]

Under the optimized condition, all 50 responses were identical:

```
Blue
```

The empirical entropy of exact strings fell to zero.

The result can therefore be described more precisely as a **collapse of realization entropy**, accompanied by a strong contraction in generated tokens.

The model did not discover an answer it previously lacked.

It ceased to materialize much of the explanation, qualification, and alternative framing surrounding an answer it was already capable of identifying.

This interpretation is more scientifically defensible than claiming that the model literally stopped “thinking through every possibility.”

The data show that it stopped **generating and accounting for** a substantial fraction of the elaboration associated with those possibilities.

5.4 Causal Limitations of Experiment Zero

The treatment was a compound intervention.

In addition to narrative structure, the optimized prompt:

- reformulated the problem's rules;
- explicitly stated that there was a single available cool primary color;
- imposed a one-word response;
- introduced an explicit fallback;
- removed permission to elaborate on ambiguity.

The control problem itself also contains a real ambiguity: green may be considered a primary color under additive RGB color models.

The optimized formulation therefore did not merely constrain the output. It partially disambiguated the underlying problem.

Experiment Zero robustly demonstrates that **a compound semantic and structural intervention can dramatically reduce output generation and latency without altering the material conclusion**.

It does not isolate the independent causal contribution of persona, narrative urgency, rule disambiguation, negative constraints, or the one-word output matrix.

That decomposition requires subsequent ablation experiments.

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6. Experiment One: Mapping the Entropic Gradient

6.1 Hypothesis

If every additional restriction were beneficial, efficiency should improve monotonically as more rules are added.

Experiment One was designed to test the opposite hypothesis:

> Beyond a certain point, instruction density may cease eliminating useful degrees of freedom and begin introducing redundancy, incompatibility, auxiliary subproblems, and additional output obligations.

The experiment used a constant contractual passage and a materially simple extraction task: identify a termination penalty of R$7,500.00.

Four prompts represented increasing levels of narrative density:

- **Level 0 — Vacuum:** open question;
- **Level 1 — Light Gravity:** legal role plus value-only response;
- **Level 2 — Formula v1.0:** strict JSON extractor with objective, scope, negative constraints, fallback, and format;
- **Level 3 — Black Hole / Overload:** hyperbolic persona, conflicting rules, underdetermined auxiliary tasks, and complex JSON output.

Thirty iterations were executed for each condition, producing 120 total observations.

Requests within each iteration were created asynchronously, while the launch order rotated systematically across conditions.

6.2 Results

Level	Description	Prompt Tokens	Completion Tokens	Total Tokens	Mean Latency	Unique Outputs
0	Vacuum	111.00	36.20	147.20	1,236.76 ms	7
1	Light Gravity	118.00	11.00	129.00	**973.96 ms**	1
2	Formula v1.0	168.00	40.83	208.83	1,305.16 ms	3
3	Black Hole / Overload	265.00	752.07	1,017.07	6,775.01 ms	30

Differences across the four levels were substantial under a non-parametric repeated-measures test:

$$
\chi^2_F(3)=75.64,
\qquad
p=2.64\times10^{-16}
$$

for latency, and

$$
\chi^2_F(3)=84.12,
\qquad
p=4.01\times10^{-18}
$$

for completion tokens.

The lowest latency did **not** occur at the condition originally labeled “Optimal Point / Formula v1.0.”

The empirical minimum occurred at **Level 1 — Light Gravity**.

This result does not weaken the proportionality hypothesis.

It falsifies the assumption that one fixed structural density is universally optimal.

For a task whose correct answer is explicit, unique, and easily localized, the complete formula introduced more structure than the problem required.

6.3 Marginal Inversion

From Level 0 to Level 1, seven additional input tokens avoided an average of 25.20 completion tokens and reduced latency by 262.80 ms:

-\frac{-25.20}{7}

3.60.
]

Each additional input token was operationally associated with the elimination of 3.60 completion tokens.

From Level 1 to Level 2, the sign reversed.

Fifty additional prompt tokens were accompanied by 29.83 additional completion tokens and 331.19 ms of additional latency:

-\frac{29.83}{50}

-0.60.
]

From Level 2 to Level 3, 97 additional input tokens accompanied an increase of 711.23 completion tokens and 5,469.85 ms of latency:

-\frac{711.23}{97}

-7.33.
]

The leverage index therefore changes sign between Levels 1 and 2.

This point represents the **semantic return inversion**: beyond it, additional specification no longer contracts total cost and instead begins to expand it.

6.4 The U-Shaped Curve

Relative to Level 0, Light Gravity reduced:

- completion tokens by 69.61%;
- total tokens by 12.36%;
- latency by 21.25%.

Relative to Level 1, Formula v1.0 increased:

- completion tokens by 271.21%;
- total tokens by 61.89%;
- latency by 34.00%.

Relative to Level 1, Overload increased:

- completion tokens by 6,736.97%;
- total tokens by 688.42%;
- latency by 595.61%.

Latency therefore exhibits the predicted qualitative shape:

initial decline → minimum → accelerated expansion.

Completion tokens showed a Pearson correlation of

[
r=0.988
]

with latency across the 120 observations.

An exploratory linear regression using only `completion_tokens` explained approximately 97.6% of the observed variance in latency.

This does not establish causality, nor does it separate generation time from queueing or network effects. It does demonstrate that, within this dataset, latency expansion was tightly associated with the expansion of generated computational work.

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7. Structural Friction: Four Distinct Mechanisms

The term **Structural Friction** should describe something more specific than a “long prompt.”

Length and conflict are not equivalent.

A long but coherent and necessary document may reduce error.

A single contradictory instruction may make a task unsatisfiable.

If each instruction (C_i) defines a set of admissible outputs, the feasible region of a request can be represented as

\bigcap_{i=1}^{n}C_i.
]

Coherent restrictions are useful when they reduce (\mathcal{F}) while preserving the correct answer.

Friction emerges when the intersection ceases to converge efficiently.

At least four mechanisms can produce this effect.

7.1 Redundancy

A redundant rule increases input cost without further reducing the feasible region.

It repeats a boundary already established or specifies behavior that the model would have adopted anyway.

Its marginal return approaches zero.

7.2 Conflict

Two rules may define incompatible sets, making

[
\mathcal{F}=\varnothing.
]

In such cases, no output can simultaneously satisfy every condition.

The model must effectively treat the request as a soft constraint-satisfaction problem: prioritize, reinterpret, ignore, or relax one or more instructions.

Recent research on instruction following under conflicting constraints shows that model performance deteriorates under such conditions and that interactions emerge between instruction position, priority, and logical compatibility.

The difficulty therefore depends not only on how many rules exist, but on the relationships among them.

7.3 Task Accretion

Adding an instruction does not necessarily constrain the original task.

It may expand it.

Requesting, in addition to the contractual penalty:

- implied inflation;
- currency conversion;
- probable cat breed;
- psychological profiling;
- legal justification

does not constitute a more precise version of the original extraction task.

It transforms a scalar target into a multi-target vector:

(\text{penalty})
]

whereas

(
\text{penalty},
\text{exchange rate},
\text{inflation},
\text{cat classification},
\text{psychological profile},
\text{legal reasoning}
).
]

The output space cannot collapse toward a single value when the specification itself re-expands that space.

7.4 Output-Matrix Inflation

A complex output matrix requires more generated tokens even when the underlying reasoning remains trivial.

If the system is instructed to produce multiple properties, justifications, and caveats, a longer response does not necessarily indicate internal inefficiency.

It may simply represent faithful execution of the specification.

Level 3 combines all four mechanisms.

It contains rules that are impossible to verify, requests calculations without supplying sufficient data, simultaneously asks the model to ignore and preserve related information, and demands an output far larger than the original target.

The model responded by attempting to resolve those gaps: declining unsupported psychological inferences, explaining why inflation or currency conversion could not be computed, adding legal caveats, and constructing large structured objects.

All 30 Level 3 outputs were textually distinct, and at least one produced formally invalid JSON.

Experiment One therefore clearly demonstrates **operational overload**, but it does not permit the entire increase to be attributed to an invisible internal struggle to reconcile constraints.

A substantial part of the observed effect comes from additional work explicitly requested by the prompt.

The scientifically defensible conclusion is therefore:

> **Conflict, redundancy, objective expansion, and output complexity can transform additional constraints into additional computational cost. The present experiment measures their combined effect.**

An isolated demonstration of a pure **constraint-reconciliation cost** will require holding the material objective, output format, and response-length ceiling constant while varying only the logical coherence of the rules.

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8. The Law of Entropic Proportionality

The original formulation of Semantic Thermodynamics treated Narrative Gravity as a relatively fixed envelope.

The new data indicate that the envelope must instead be adaptive.

What should remain invariant is not the amount of structure, but the correspondence between structural pressure and problem complexity.

The **entropic mass** of a task may be represented as a vector

\begin{bmatrix}
e_a\
e_d\
e_r\
e_s\
e_u\
e_f
\end{bmatrix},
]

where:

- (e_a): ambiguity among semantically distinct answers;
- (e_d): density of distractors in the input;
- (e_r): reasoning depth required;
- (e_s): complexity of the output schema;
- (e_u): evidentiary insufficiency or uncertainty;
- (e_f): expected cost of failure.

Constraint density can similarly be represented by

\begin{bmatrix}
g_\Psi\
g_\Omega\
g_{\oint}\
g_{\bar X}\
g_M\
g_{S_0}
\end{bmatrix}.
]

The **Entropic Proportionality Tensor** (\Lambda) can then be defined as the operator relating task properties to appropriate constraint intensities:

\Lambda_{\theta,s,\tau}
(\mathbf{e}),
]

where:

- (\theta) identifies the model;
- (s) identifies the serving regime;
- (\tau) identifies the required quality threshold.

The optimal narrative density is defined as

\underset{\mathbf{g}}{\operatorname{arg,min}};
J(\mathbf{g};\mathbf{e})
]

subject to

[
P(\text{correct answer}\mid\mathbf{g},\mathbf{e})
\geq \tau
]

and, when applicable,

[
P(\text{structurally valid output}\mid\mathbf{g},\mathbf{e})

\geq \sigma.
]

Structural Friction can then be operationalized as excess cost relative to the optimum:

J(\mathbf{g};\mathbf{e})

J(\mathbf{g}^{*};\mathbf{e}).
]

By definition,

[
F_s\geq0.
]

The **Black Hole** regime is therefore not a mysterious state.

It is the region in which the marginal cost of an additional constraint exceeds its marginal capacity to reduce error or unnecessary generation:

[
\frac{\partial J}{\partial |\mathbf{g}|}

]

A simple scalar approximation is

J_0

a(e)g
+
b(e)g^2
+
c,\chi(g),
]

where:

- the negative linear term represents the initial benefit of structure;
- the quadratic term represents over-specification;
- (\chi(g)) represents logical conflict among restrictions.

In the absence of conflict, the minimum occurs at

\frac{a(e)}{2b(e)}.
]

For simple problems, (a(e)) is small: there is little unnecessary uncertainty to remove, and (g^{*}) shifts leftward.

For ambiguous, long-context, or high-risk problems, the initial benefit of restriction may be substantially larger, shifting the optimum to the right.

This produces a falsifiable formulation of the **Law of Entropic Proportionality**:

> **Holding model, serving regime, and quality threshold constant, the optimal density of semantic constraint should increase with the entropic mass of the task; beyond that optimum, the marginal cost of additional structure becomes positive.**

The experiments reported here provide initial evidence for the existence of an optimum and for its displacement toward lighter constraint under low-entropy tasks.

They are not yet sufficient to estimate a universal form of (\Lambda).

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9. Narrative as a Semantic Compiler

A conventional compiler transforms a formal specification into an executable representation.

Operational narrative performs an analogous function at a probabilistic level: it transforms broad human intent into a more concentrated distribution of admissible continuations.

The Persona selects a policy family.

The Objective defines the terminal state.

The Scope defines admissible evidence.

Negative Constraints eliminate destinations.

The Matrix determines the grammar of materialization.

The Ground State provides a safe terminal fallback.

Narrative does not replace constrained decoding.

The two mechanisms operate at different layers.

A grammar may prevent the production of syntactically invalid JSON.

It cannot, by itself, determine whether a field should contain a value, `null`, a refusal, or an unsupported inference.

Narrative constrains **meaning**.

Constrained decoding constrains **syntax**.

Reliable systems may require both.

Nor does the framework imply that dramatic personas are inherently superior.

Experiment One points in the opposite direction.

For a trivial extraction task, the instruction:

> “You are a lawyer. Extract the termination penalty and respond only with the value.”

outperformed the complete Formula v1.0.

The relevant quantity is therefore not rhetorical intensity, but **useful restrictive information per token**.

The best narrative is not the longest, most solemn, or most detailed.

It is the one in which every component removes a real uncertainty in the task.

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10. In What Sense Is This Quantity Thermodynamic?

Landauer's principle established a fundamental connection between logically irreversible information processing and physical dissipation, reinforcing the broader principle that information does not exist independently of a physical substrate.

That relationship does **not**, however, justify treating every reduction in token count as a direct measurement of heat.

Landauer's bound concerns fundamental physical limits of information operations.

The experiments reported here measure API-level telemetry and response time in an opaque production infrastructure many abstraction layers above that limit.

Narrative may therefore be described as an **operational thermodynamic quantity** only in a restricted sense.

First, it modifies a probabilistic distribution over computationally realized output states.

Second, that modification changes measurable material quantities such as generated tokens, execution time, and monetary cost.

Third, the system exhibits an efficiency regime, an optimum, and a region of increasing dissipation or friction.

The qualifier **operational** is essential.

Until electrical consumption, accelerator utilization, execution cycles, or energy per request are directly measured, the observed reductions cannot be claimed to correspond proportionally to reductions in physical heat or energy.

Nor do these experiments establish narrative as a new fundamental force of nature.

They support a narrower and more concrete proposition:

> **Structured meaning, when encoded as an instruction, changes the physical work performed by a machine while producing a response.**

No mysticism is required.

Every prompt is information represented on physical hardware.

Every inference is computation executed by a physical system.

Any systematic modification in the amount or duration of generated computation has material consequences.

The scientific question is therefore to quantify the transfer function between semantic structure and computational cost.

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11. Implications for AI Infrastructure

The enterprise implication of this theory is not to wrap every request in the complete Semantic Thermodynamics formula.

It is to construct a **semantic-density router**.

Before producing the final instruction, a system could estimate the entropic mass of the task:

- number of plausible answers;
- distribution of relevant evidence;
- distractor density;
- output complexity;
- reasoning depth;
- uncertainty;
- cost of error.

The system would then select the smallest set of constraints capable of meeting the required quality threshold.

A direct field-extraction request may require only an objective and output format.

A contract containing competing clauses may require a specialized persona, explicit scope, negative constraints, and a fallback state.

A task containing mutually incompatible objectives may not require a denser prompt at all.

It may require decomposition into separate operations.

Such routing can be represented as a policy:

$$
\pi: \mathbf{e} \longrightarrow \mathbf{g}^{*}.
$$

The net financial effect of a narrative intervention can be expressed as

$$
c_p\Delta T_p + c_c\Delta T_c + c_\ell\Delta L + c_e\Delta \mathcal{E}.
$$

The intervention is economically efficient when

$$
\Delta C<0.
$$

In Experiment Zero, the additional cost of 48 input tokens was compensated by a mean saving of 242.38 completion tokens.

Across the first gradient in Experiment One, seven additional input tokens saved 25.20 completion tokens.

Across subsequent gradients, the return became negative.

This demonstrates why prompt length in isolation is an inadequate optimization metric.

The correct object of optimization is the **complete inference cycle**.

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12. Limitations

The results are strong as a proof of concept but do not support unrestricted generalization.

First, each experiment uses only one material task.

Experiment Zero includes a color problem with genuine interpretive ambiguity.

Experiment One uses an extremely simple extraction task.

The shape of the curve remains unknown for programming, mathematics, document retrieval, creative generation, classification, agent planning, and long-horizon reasoning.

Second, the experimental scripts were configured with a default model while allowing replacement through an environment variable, and the resulting CSV files do not record the effectively resolved model identifier, model snapshot, system fingerprint, reasoning effort, or service tier.

Third, latency includes network transmission, queueing, and provider-side conditions.

Paired execution and rotating request order reduce some of this noise but cannot eliminate it.

Fourth, `completion_tokens` was not decomposed into visible output tokens, reasoning tokens, and other internally generated components.

This prevents direct identification of which fraction of the observed contraction occurred in non-visible reasoning.

Fifth, the treatments modify several properties simultaneously.

Experiment Zero combines disambiguation, persona, narrative urgency, output restriction, and fallback behavior.

Experiment One combines density, conflict, task expansion, and output complexity.

Sixth, textual diversity is not equivalent to semantic entropy.

Outputs may differ lexically while belonging to the same semantic equivalence class, as demonstrated by Experiment Zero.

Future studies should cluster outputs by semantic equivalence before calculating answer entropy.

Finally, no direct measurement of power consumption, accelerator occupancy, floating-point operations, or energy was performed.

At the present stage, **thermodynamics** should therefore be understood as an effective theory of computational cost and probability distributions, not as a demonstrated one-to-one equivalence with fundamental thermal quantities.

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13. Required Experimental Program

The next stage should move beyond compound interventions and adopt a factorial design.

Each of the six components—

[
\Psi,;
\vec{\Omega},;
\oint,;
\bar X,;
M,;
S_0
]

—should be independently activated and removed.

A full factorial design would allow estimation of main effects and interactions.

A fractional factorial design could reduce the number of API calls while preserving the most informative combinations.

A dedicated conflict experiment should preserve:

- identical source text;
- identical material objective;
- identical output schema;
- identical response-length ceiling;
- approximately equal prompt length.

The only manipulated variable should be the logical relationship among rules:

coherent, redundant, or contradictory.

Under such a design, any increase in internal generation could be attributed with substantially greater confidence to constraint reconciliation rather than expansion of the requested task.

Future telemetry should include

[
T_{\text{input}},
\quad
T_{\text{visible}},
\quad
T_{\text{reasoning}},
\quad
TTFT,
\quad
TPOT,
\quad
L_{\text{total}},
]

together with:

- resolved model identifier;
- model snapshot;
- system fingerprint;
- cache status;
- reasoning-effort setting;
- termination reason;
- structural-validity score.

To establish the genuinely physical dimension of the theory, the protocol should subsequently be reproduced using open-weight models running on controlled hardware, with direct sampling of power and energy consumption per request.

Only then will it be possible to test whether contractions in tokens and latency correspond systematically to contractions in joules.

The experiment must also vary input entropy itself.

Rather than relying on four intuitively labeled prompt levels, future studies should construct tasks with controlled variation in:

- number of plausible answers;
- proportion of distractors;
- degree of contradiction;
- amount of missing evidence;
- reasoning depth;
- output dimensionality.

The function

[
J(g;e)
]

could then be estimated across multiple values of (e), allowing direct testing of whether the optimal point shifts as predicted by (\Lambda).

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14. Conclusion

Experiment Zero demonstrated that a semantically constrained formulation can tolerate a moderate increase in prompt length while producing a far larger contraction in generated output, total token expenditure, and end-to-end latency.

The material answer remained unchanged.

The number of textual realizations collapsed from fifty to one.

Experiment One demonstrated that this benefit does not increase indefinitely.

For a low-entropy task, a minimal and directly targeted instruction outperformed both the unconstrained baseline and the full formula.

When additional constraints began introducing conflict, auxiliary fields, unsupported inferences, and secondary objectives, computational cost increased dramatically.

The central result is therefore neither that **long prompts are bad** nor that **dramatic narratives make models more intelligent**.

It is that:

> **Constraints have marginal returns.**

Before the optimum, they remove degrees of freedom that the model would otherwise need to resolve or materialize.

Beyond the optimum, they consume context, multiply obligations, create incompatibilities, and re-expand the very output space they were intended to compress.

Narrative thus emerges as a form of **semantic compilation**.

It converts intention into boundaries, boundaries into probability distributions, and probability distributions into computational work.

Its value is not determined by the number of constraints imposed, but by the correspondence between each constraint and a genuine uncertainty in the task.

The Law of Entropic Proportionality can therefore be summarized in a single proposition:

> **Optimal structure is not maximal. It is proportional.**

To constrain infinity is not to construct the densest possible prison around a model.

It is to discover the smallest boundary capable of transforming a field of possibilities into an action that is correct, stable, and economically efficient.

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